

AZIM RAHAT

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EDUCATION

The City College of New York, The City University of New York

Graduation December 2026

B.S. Computer Science | 3.7 GPA

- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Database Systems, Systems Design, Software Engineering, Computer Architecture, Statistics, Image Processing, Machine Learning

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, Java, SQL, CUDA

Frameworks: React, Node.js, Express, Next.js, Flask, Django, FastAPI, NestJS, Tailwind CSS

Libraries: Scikit-learn, Pandas, TensorFlow, Selenium, GraphQL, Dask

Databases/Tools: PostgreSQL, MongoDB, Redis, AWS, Docker, Kubernetes, Git, Kafka, Datadog

EXPERIENCE

OnePay

June 2026 – Aug 2026

Software Engineer Intern

New York, NY

- Built a voice-agent QA platform replacing a **\$60k/yr** vendor tool, streaming live audio over ElevenLabs WebSockets to run LLM-simulated test scenarios against a customer-support agent
- Increased voice-agent eval throughput **10x**, automating manual phone-call testing and post-call grading via an LLM judge
- Designed an event-driven backend with **SQS** queuing to a worker pool, **DynamoDB** for run state, and **S3** for recordings; published events to **Kafka** with **Datadog** monitoring for latency and pipeline health
- Engineered simulated caller with configurable audio behavior (accented speech, background noise, interruptions), surfacing **production edge cases** that drove guardrail and tool-call fixes
- Automated eval test-case creation from production conversations, replaying each interaction's tool calls as saved mocks under existing PII controls and cutting creation time from **~5 minutes to seconds**

STEMKasa

May 2024 – May 2025

Software Engineer Intern

New York, NY

- Built Flask/Express APIs using **GPT-4** and **Whisper** for voice-based Q&A, practice-problem generation, and grading
- Reduced average API response time by **22%** by introducing a Redis caching layer in front of frequently-accessed MongoDB queries, improving concurrent request handling under load
- Authored **50+** unit and e2e tests integrated into a CI/CD pipeline, catching regressions and reducing pre-deployment failures
- Shipped **10+** reusable React components (forms, dashboards, inputs) adopted team-wide, accelerating frontend delivery

PROJECTS

Jumble | WebSockets, Redis, Docker, AWS, React, TypeScript, Express, PostgreSQL

[live site](#) | [github](#)

- Engineered a real-time multiplayer platform supporting 8-player lobbies, room codes, and synchronized scoring/timer logic through **WebSockets** and Redis caching for responsive state updates
- Deployed scalable backend with **Dockerized** services (backend, PostgreSQL, Redis) and automated database migrations, integrating a GitHub Actions CI/CD pipeline for reliable deployments on AWS (RDS, EC2)
- Grew to **110+** active users post-launch, validating system reliability under concurrent real-time load

GPU-Accelerated Recommendation System | RAPIDS (cuDF, cuML), CUDA, Python, Dask, Pandas, FastAPI

[github](#)

- Built a GPU-accelerated recommender pipeline over **571M** Amazon reviews and **600k+** items using NVIDIA RAPIDS, achieving **5.15x** end-to-end speedup over CPU baseline
- Engineered a custom memory layout with **NVIDIA RMM** and ID remapping, fitting a **26 GB** sparse matrix on a **24 GB** GPU and reclaiming **4-6 GB** of VRAM
- Trained a matrix factorization model on **31M** interactions with **59x** faster GPU training, served at **1.4ms** latency via FastAPI

AWARDS & EXTRACURRICULARS

- **Dean's List** (Fall 2022 – Spring 2025), **Grove School of Engineering Scholarship** (2023 – 2025)
- **Board Member, ACM (CCNY)** – Organized coding workshops, technical talks, and peer mentoring sessions for CS students
- **Google Software Engineering Fellowship** – selected from **350+** applicants for mentorship on system design and algorithms